

Erdős Problem #1039

EP-1039

L1 - TRACTABLE

PARTIALLY SOLVED

Combinatorics

ORIGINAL PROBLEM STATEMENT

Let $f(z) = \prod_{i=1}^n (z - z_i) \in \mathbb{C}[z]$ with $|z_i| \leq 1$ for all i . Let $\rho(f)$ be the radius of the largest disc which is contained in $\{z : |f(z)| < 1\}$.

Determine the behaviour of $\rho(f)$. In particular, is it always true that $\rho(f) \gg 1/n$?

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Archive: <https://www.unsolvedmath.com/problems?difficulty=1>

Dataset: <https://huggingface.co/datasets/ulamai/UnsolvedMath>